8. Depth First Search (DFS)

Aim:

To write a Python program to implement the Depth First Search (DFS) algorithm.

Algorithm:

1. Start the program.
2. Define a graph using an adjacency list.
3. Initialize the starting node.
4. Mark the node as visited.
5. Display the node.
6. Visit an unvisited adjacent node.
7. Continue recursively until no unvisited adjacent nodes remain.
8. Backtrack to the previous node.
9. Repeat until all reachable nodes are visited.
10. Stop.

INPUT

```
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

def dfs(node, visited):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbor in graph[node]:
            dfs(neighbor, visited)

dfs('A', set())
```

OUPUT

Result:

Thus, the Python program to implement Depth First Search was executed successfully.